

Simultaneous Augmentation of Textures and Deformation Based on Dynamic Projection Mapping

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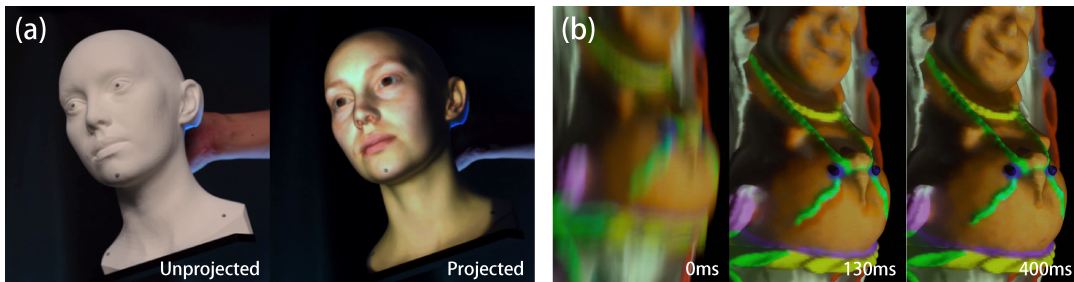


Figure 1: Dynamic projection mapping by the proposed system. (a) Texture overwriting by the projection. (b) Virtual elasticity caused by a perceptual illusion. The middle of the figure shows intentional displacement of luminance inducing virtual deformation on a rigid body. This illusion is effective when viewed through videos. Please refer to our supplemental video. 3D models are from Wikihuman Project [Ma et al. 2007] and the Stanford Computer Graphics Laboratory.

ABSTRACT

In this paper, we exploit human perception characteristics and dynamic projection mapping techniques and realize overwriting of both textures and deformation of a real object. To keep the projection following a moving object and induce deformation illusion, we developed a 1000 fps projector-camera system and demonstrated augmentation of the real world. In the demonstration, the audience will see a plaster figure turning into a colorful and flabby object.

ACM Reference Format:

Leo Miyashita, Kentaro Fukamizu, and Masatoshi Ishikawa. 2021. Simultaneous Augmentation of Textures and Deformation Based on Dynamic Projection Mapping. In *SIGGRAPH Asia 2021 Emerging Technologies (SA '21 Emerging Technologies)*, December 14-17, 2021. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3476122.3484838>

1 INTRODUCTION

An appearance of a real object is determined by the shape, material, and lighting, and the human cannot accurately recognize which factor contributes to the visual representation. A projection mapping technique utilizes this insight and enables visual augmentation of the real world without any equipment like specialized goggles. Furthermore, a dynamic projection mapping (DPM) technique that can target dynamically moving or deforming objects has recently been proposed, and new computer-based visual representation is being

created. For example, Narita et al. developed occlusion-robust markers called DDCM and a texture overwriting system for non-rigid surface such as paper and a T-shirt [Narita et al. 2017]. Additionally, Miyashita et al. proposed image-based dynamic projection mapping to overwrite materials without any markers and shape models [Miyashita et al. 2018].

However, most of the conventional researches focus only on overwriting of the texture or material, namely, optical characteristics. On the other hand, appearance is influenced also by the shape, and so lighting should be able to augment the shape by exploiting human perception. In fact, Fukamizu et al. succeeded in embodying virtual deformation and overwrote elasticity, namely, mechanical characteristics by projecting designed grayscale images onto a color-textured object [Fukamizu et al. 2020]. Nevertheless, this research conversely focuses only on overwriting of the shape and motion.

In this project, we combine the texture and deformation overwriting methods, and demonstrate comprehensive augmentation of appearance. This proposal involves academic questions including the validity of the simultaneous overwriting method, and dependencies among visual effects, dynamic range, and rendering parameters for deformation illusion. Through this project, we seek the potential of visual augmentation by DPM.

In our demonstration, the audience will see the real object with unrealistic visual effects such as a video-textured plaster figure shaking like a jelly in response to the motion. The motion of the target is fed back to the projection at so high speed that humans cannot notice the delay, and so the virtual texture and deformation will be perceived as the original characters. For more details of the experience, please refer to our supplemental video.

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SA '21 Emerging Technologies, December 14-17, 2021, Tokyo, Japan

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ACM ISBN 978-1-4503-8685-2/21/12.

<https://doi.org/10.1145/3476122.3484838>

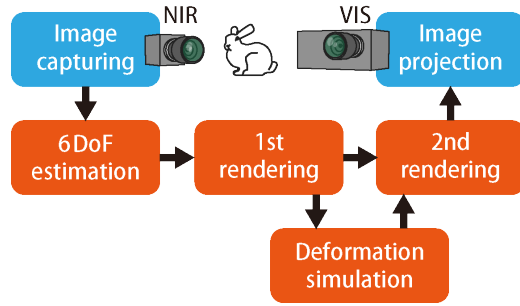


Figure 2: The proposed workflow.

2 AUGMENTATION OF TEXTURES AND DEFORMATION BY DPM

To realize the augmentation of a texture and deformation, we propose the workflow shown in Fig. 2. First, markers on the target surface are detected in the near-infrared (NIR) region and the pose, position, and velocity of the target are estimated. Note that the marker layout and the shape of the target are known. Second, the virtual-textured appearance of the target is rendered based on the pose and position, and so the rendered image is supposed to perfectly fit the outline of the target in this first rendering. Third, the virtually deformed shape of the target is computed based on the velocity and physical simulation, and the geometry protruded from the 2D outline of the target in the first rendered image is corrected by blending the original geometry [Fukamizu et al. 2020]. This correction works to keep the projection only inside the target and enhance the immersive experience. Finally, the second rendering is performed as the following equation and the result I_{result} is projected onto the target object in the visible (VIS) light region.

$$I_{\text{result}} = I(M) + \rho C(I(M') - I(M)) \quad (1)$$

Here, $I(\cdot)$ denotes a rendered image with the original shape M or the deformed shape M' , and $C(\cdot)$, ρ respectively stand for a color conversion function and an amplitude term for deformation illusion. In this equation, the first term of the right side represents the virtual textures and materials, and the last term causes displacement of luminance inducing virtual deformation [Kawabe et al. 2016].

The actual system shown in Fig. 3 is composed of a high-speed camera (monochrome, 640×480 @1000 fps), a high-speed projector (24bit-color, 1024×672 @1000 fps), a cold mirror, and infrared ring light. The workflow was processed by a CPU (Intel, i7-6950X @3.0GHz) and GPU (NVIDIA, Quadro RTX 8000), and latency from capturing to projection was about 6 ms. This latency is low enough to be unnoticeable for humans [Ng et al. 2012]. Besides, we set the amplitude term as $\rho = 0.5$ in the experiments.

Figure 1 shows the actual projection by the proposed system. Both targets are originally white as shown on the left of figure (a), and all color textures are based on the projection. The left of figure (b) shows that the projection followed the moving target with low latency and caused motion blur. After the target stopped the movement, the displacement of luminance clearly appeared in

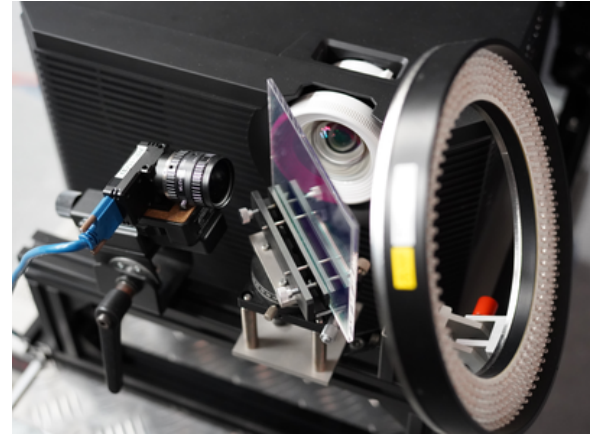


Figure 3: The proposed dynamic projection mapping system.

the middle of figure (b). For more results and deformation illusion, please refer to our supplemental video.

3 CONCLUSION

In this paper, we proposed the method to simultaneously augment textures and deformation of real objects by dynamic projection mapping techniques. The high-speed and low-latency visual feedback enables virtual overwriting of visual representation and connects the real world and a virtual world seamlessly. These visual effects in real life are possible since the recipients are humans, and such distinctive advantages are expected to be utilized increasingly for new visual representation in the future.

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